Problem 2: Matrix Diagonal Sum (easy)

Problem Statement

Given a square matrix (2D array), calculate the sum of its two diagonals.

The two diagonals in consideration are the primary diagonal that spans from the top-left to the bottom-right and the secondary diagonal that spans from top-right to bottom-left. If a number is part of both diagonals (which occurs only for odd-sized matrices), it should be counted only once in the sum.

Examples

1. **Example 1**:
   - **Input**:

     ```
     [[1,2,3],
      [4,5,6],
      [7,8,9]]
     ```

   - **Expected Output**: 25
   - **Justification**: Summing up the two diagonals (1+5+9+3+7), we get 25. Please note that the element at [1][1] = 5 is counted only once.
2. **Example 2**:
   - **Input**:

     ```
     [[1,0],
      [0,1]]
     ```

   - **Expected Output**: 2
   - **Justification**: The sum of the two diagonals is 1+1 = 2.
3. **Example 3**:
   - **Input**:

     ```
     [[5]]
     ```

   - **Expected Output**: 5
   - **Justification**: Since there's only one element, it is the sum itself.

Try it yourself

Try solving this question here:

Java

```java
public class Solution {
    public static int diagonalSum(int[][] mat) {
        int totalSum = 0;  // Initialize the total sum
        // ToDo: Write Your Code Here.
        return totalSum;  // Return the calculated total sum
    }
}
```

Test Case | Run Code Result | Stdout

[[1,2],[3,4]]

Use Example Testcases | Run | Submit